

EA-0058 - AQELYN Engineering Archive

EA-0058 — Development & Coding Standards

Brand: AQELYN

Subtitle: Normative coding standard for AQELYN Platform implementation

Status: FULL_COMPLETE

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Executive Summary

Define the mandatory coding, review, security, testing, documentation, and release rules that all implementation work must follow.

This document is part of the AQELYN / AQELYN Architecture Baseline v1.0 and is normative for implementation. It supports the product principles: Explain Before You Recommend, Simplicity First, Evidence Before Opinion, Human-Centered Security, and Expert Depth on Demand.

Scope Domains

- Repository and module boundaries
- Python backend standards
- TypeScript frontend standards
- API and schema conventions
- Secure coding requirements
- Testing and quality gates
- Git workflow and release discipline
- Definition of Done

01. Engineering Authority

This section defines the mandatory rules, constraints, and implementation expectations for **Engineering Authority** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-01-01:** The implementation shall apply engineering authority consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-01-02:** The implementation shall apply engineering authority consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-01-03:** The implementation shall apply engineering authority consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-01-04:** The implementation shall apply engineering authority consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-01-05:** The implementation shall apply engineering authority consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Engineering Authority is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Engineering Authority is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Engineering Authority is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Engineering Authority is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Engineering Authority, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

02. Repository Compliance

This section defines the mandatory rules, constraints, and implementation expectations for **Repository Compliance** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-02-01:** The implementation shall apply repository compliance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-02-02:** The implementation shall apply repository compliance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-02-03:** The implementation shall apply repository compliance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-02-04:** The implementation shall apply repository compliance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-02-05:** The implementation shall apply repository compliance consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Repository Compliance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Repository Compliance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Repository Compliance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Repository Compliance is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Repository Compliance, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

03. Language Standards

This section defines the mandatory rules, constraints, and implementation expectations for **Language Standards** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-03-01:** The implementation shall apply language standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-03-02:** The implementation shall apply language standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-03-03:** The implementation shall apply language standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-03-04:** The implementation shall apply language standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-03-05:** The implementation shall apply language standards consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Language Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Language Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Language Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Language Standards is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Language Standards, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

04. Backend Standards

This section defines the mandatory rules, constraints, and implementation expectations for **Backend Standards** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-04-01:** The implementation shall apply backend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-04-02:** The implementation shall apply backend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-04-03:** The implementation shall apply backend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-04-04:** The implementation shall apply backend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-04-05:** The implementation shall apply backend standards consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Backend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Backend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Backend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Backend Standards is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Backend Standards, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

05. Frontend Standards

This section defines the mandatory rules, constraints, and implementation expectations for **Frontend Standards** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-05-01:** The implementation shall apply frontend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-05-02:** The implementation shall apply frontend standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-05-03:** The implementation shall apply frontend standards consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0058-REQ-05-04:**** The implementation shall apply frontend standards consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-05-05:**** The implementation shall apply frontend standards consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Frontend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Frontend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Frontend Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Frontend Standards is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Frontend Standards, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

06. API Conventions

This section defines the mandatory rules, constraints, and implementation expectations for ****API Conventions**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0058-REQ-06-01:**** The implementation shall apply api conventions consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-06-02:**** The implementation shall apply api conventions consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-06-03:**** The implementation shall apply api conventions consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-06-04:**** The implementation shall apply api conventions consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-06-05:**** The implementation shall apply api conventions consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.

- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: API Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: API Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: API Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: API Conventions is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to API Conventions, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

07. Data Model Conventions

This section defines the mandatory rules, constraints, and implementation expectations for **Data Model Conventions** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-07-01:** The implementation shall apply data model conventions consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-07-02:** The implementation shall apply data model conventions consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-07-03:** The implementation shall apply data model conventions consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-07-04:** The implementation shall apply data model conventions consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-07-05:** The implementation shall apply data model conventions consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Data Model Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Data Model Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Data Model Conventions is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Data Model Conventions is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Data Model Conventions, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

08. Configuration Rules

This section defines the mandatory rules, constraints, and implementation expectations for **Configuration Rules** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-08-01:** The implementation shall apply configuration rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-08-02:** The implementation shall apply configuration rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-08-03:** The implementation shall apply configuration rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-08-04:** The implementation shall apply configuration rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-08-05:** The implementation shall apply configuration rules consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Configuration Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Configuration Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Configuration Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Configuration Rules is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Configuration Rules, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

09. Logging Rules

This section defines the mandatory rules, constraints, and implementation expectations for **Logging Rules** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-09-01:** The implementation shall apply logging rules consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0058-REQ-09-02:**** The implementation shall apply logging rules consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-09-03:**** The implementation shall apply logging rules consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-09-04:**** The implementation shall apply logging rules consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-09-05:**** The implementation shall apply logging rules consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Logging Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Logging Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Logging Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Logging Rules is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Logging Rules, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

10. Error Handling

This section defines the mandatory rules, constraints, and implementation expectations for ****Error Handling**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0058-REQ-10-01:**** The implementation shall apply error handling consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-10-02:**** The implementation shall apply error handling consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-10-03:**** The implementation shall apply error handling consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-10-04:**** The implementation shall apply error handling consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-10-05:**** The implementation shall apply error handling consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Error Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Error Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Error Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Error Handling is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Error Handling, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

11. Security Coding Rules

This section defines the mandatory rules, constraints, and implementation expectations for **Security Coding Rules** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-11-01:** The implementation shall apply security coding rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-11-02:** The implementation shall apply security coding rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-11-03:** The implementation shall apply security coding rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-11-04:** The implementation shall apply security coding rules consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-11-05:** The implementation shall apply security coding rules consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Security Coding Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Security Coding Rules is implemented, tested, documented, and linked to the traceability matrix.

- Acceptance item 3: Security Coding Rules is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Security Coding Rules is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Security Coding Rules, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

12. Dependency Governance

This section defines the mandatory rules, constraints, and implementation expectations for **Dependency Governance** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-12-01:** The implementation shall apply dependency governance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-12-02:** The implementation shall apply dependency governance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-12-03:** The implementation shall apply dependency governance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-12-04:** The implementation shall apply dependency governance consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-12-05:** The implementation shall apply dependency governance consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Dependency Governance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Dependency Governance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Dependency Governance is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Dependency Governance is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Dependency Governance, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

13. Secrets Handling

This section defines the mandatory rules, constraints, and implementation expectations for **Secrets Handling** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-13-01:** The implementation shall apply secrets handling consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-13-02:** The implementation shall apply secrets handling consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-13-03:** The implementation shall apply secrets handling consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-13-04:** The implementation shall apply secrets handling consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-13-05:** The implementation shall apply secrets handling consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Secrets Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Secrets Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Secrets Handling is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Secrets Handling is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Secrets Handling, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

14. Testing Standards

This section defines the mandatory rules, constraints, and implementation expectations for **Testing Standards** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-14-01:** The implementation shall apply testing standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-14-02:** The implementation shall apply testing standards consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-14-03:** The implementation shall apply testing standards consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0058-REQ-14-04:**** The implementation shall apply testing standards consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-14-05:**** The implementation shall apply testing standards consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Testing Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Testing Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Testing Standards is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Testing Standards is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Testing Standards, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

15. Static Analysis

This section defines the mandatory rules, constraints, and implementation expectations for ****Static Analysis**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0058-REQ-15-01:**** The implementation shall apply static analysis consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-15-02:**** The implementation shall apply static analysis consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-15-03:**** The implementation shall apply static analysis consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-15-04:**** The implementation shall apply static analysis consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-15-05:**** The implementation shall apply static analysis consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.

- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Static Analysis is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Static Analysis is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Static Analysis is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Static Analysis is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Static Analysis, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

16. Git Workflow

This section defines the mandatory rules, constraints, and implementation expectations for **Git Workflow** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-16-01:** The implementation shall apply git workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-16-02:** The implementation shall apply git workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-16-03:** The implementation shall apply git workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-16-04:** The implementation shall apply git workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-16-05:** The implementation shall apply git workflow consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Git Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Git Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Git Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Git Workflow is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Git Workflow, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

17. Pull Request Workflow

This section defines the mandatory rules, constraints, and implementation expectations for **Pull Request Workflow** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-17-01:** The implementation shall apply pull request workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-17-02:** The implementation shall apply pull request workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-17-03:** The implementation shall apply pull request workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-17-04:** The implementation shall apply pull request workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-17-05:** The implementation shall apply pull request workflow consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Pull Request Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Pull Request Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Pull Request Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Pull Request Workflow is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Pull Request Workflow, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

18. Code Review

This section defines the mandatory rules, constraints, and implementation expectations for **Code Review** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-18-01:** The implementation shall apply code review consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0058-REQ-18-02:**** The implementation shall apply code review consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-18-03:**** The implementation shall apply code review consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-18-04:**** The implementation shall apply code review consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-18-05:**** The implementation shall apply code review consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Code Review is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Code Review is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Code Review is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Code Review is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Code Review, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

19. Release Quality Gates

This section defines the mandatory rules, constraints, and implementation expectations for ****Release Quality Gates**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0058-REQ-19-01:**** The implementation shall apply release quality gates consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-19-02:**** The implementation shall apply release quality gates consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-19-03:**** The implementation shall apply release quality gates consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-19-04:**** The implementation shall apply release quality gates consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0058-REQ-19-05:**** The implementation shall apply release quality gates consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Release Quality Gates is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Release Quality Gates is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Release Quality Gates is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Release Quality Gates is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Release Quality Gates, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

20. Definition of Done

This section defines the mandatory rules, constraints, and implementation expectations for **Definition of Done** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0058-REQ-20-01:** The implementation shall apply definition of done consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-20-02:** The implementation shall apply definition of done consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-20-03:** The implementation shall apply definition of done consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-20-04:** The implementation shall apply definition of done consistently across all affected modules and maintain traceability to source requirements.
- **EA-0058-REQ-20-05:** The implementation shall apply definition of done consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Definition of Done is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Definition of Done is implemented, tested, documented, and linked to the traceability matrix.

- Acceptance item 3: Definition of Done is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Definition of Done is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Definition of Done, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

Requirements Matrix

Requirement ID	Requirement	Priority	Verification
EA-0058-REQ-01	Implement Engineering Authority controls	Mandatory	Review + Tests
EA-0058-REQ-02	Implement Repository Compliance controls	Mandatory	Review + Tests
EA-0058-REQ-03	Implement Language Standards controls	Mandatory	Review + Tests
EA-0058-REQ-04	Implement Backend Standards controls	Mandatory	Review + Tests
EA-0058-REQ-05	Implement Frontend Standards controls	Mandatory	Review + Tests
EA-0058-REQ-06	Implement API Conventions controls	Mandatory	Review + Tests
EA-0058-REQ-07	Implement Data Model Conventions controls	Mandatory	Review + Tests
EA-0058-REQ-08	Implement Configuration Rules controls	Mandatory	Review + Tests
EA-0058-REQ-09	Implement Logging Rules controls	Mandatory	Review + Tests
EA-0058-REQ-10	Implement Error Handling controls	Mandatory	Review + Tests
EA-0058-REQ-11	Implement Security Coding Rules controls	Mandatory	Review + Tests
EA-0058-REQ-12	Implement Dependency Governance controls	Mandatory	Review + Tests
EA-0058-REQ-13	Implement Secrets Handling controls	Mandatory	Review + Tests
EA-0058-REQ-14	Implement Testing Standards controls	Mandatory	Review + Tests
EA-0058-REQ-15	Implement Static Analysis controls	Mandatory	Review + Tests
EA-0058-REQ-16	Implement Git Workflow controls	Mandatory	Review + Tests
EA-0058-REQ-17	Implement Pull Request Workflow controls	Mandatory	Review + Tests
EA-0058-REQ-18	Implement Code Review controls	Mandatory	Review + Tests
EA-0058-REQ-19	Implement Release Quality Gates controls	Mandatory	Review + Tests
EA-0058-REQ-20	Implement Definition of Done controls	Mandatory	Review + Tests

Traceability Matrix

Requirement	Source	Implementation Target	Verification Evidence
EA-0058-REQ-01	EA-0058 Section 01	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-02	EA-0058 Section 02	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-03	EA-0058 Section 03	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-04	EA-0058 Section 04	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-05	EA-0058 Section 05	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-06	EA-0058 Section 06	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-07	EA-0058 Section 07	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-08	EA-0058 Section 08	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-09	EA-0058 Section 09	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-10	EA-0058 Section 10	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-11	EA-0058 Section 11	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist

EA-0058-REQ-12	EA-0058 Section 12	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-13	EA-0058 Section 13	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-14	EA-0058 Section 14	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-15	EA-0058 Section 15	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-16	EA-0058 Section 16	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-17	EA-0058 Section 17	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-18	EA-0058 Section 18	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-19	EA-0058 Section 19	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0058-REQ-20	EA-0058 Section 20	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist

Engineering Journal

- 2026-07-09: EA-0058 created as part of the final architecture completion set EA-0058 through EA-0061.
- Decision: AQELYN branding is applied while AQELYN remains the internal engineering codename.
- Decision: this package is normative for coding readiness and Codex execution.

Final Review

The package is approved for implementation baseline use when all files are present, hashes are generated, diagrams render, and PDF/HTML outputs correspond to the Master Markdown.